RedShift Aerospace Propulsion Engineering Team

Laminar propulsion model

Using Euler-Lagrange equations

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Model premises:

-The model will aim to determine the initial mass of rocket fuel based on the following physical aspect that we aim to obtain:

$$\begin{cases} The\ propultion\ flow\ will\ be\ laminar \\ The\ target\ height\ is\ 9000m = 9km \\ Air\ drag\ and\ rotation\ of\ the\ solid\ rigit\ will\ be\ included \end{cases}$$

-We intend to maintain the stability of the rocket, but it would be helpful to resolve the problem for a point like particle, so we have found a way to incorporate both aspects using the following model:

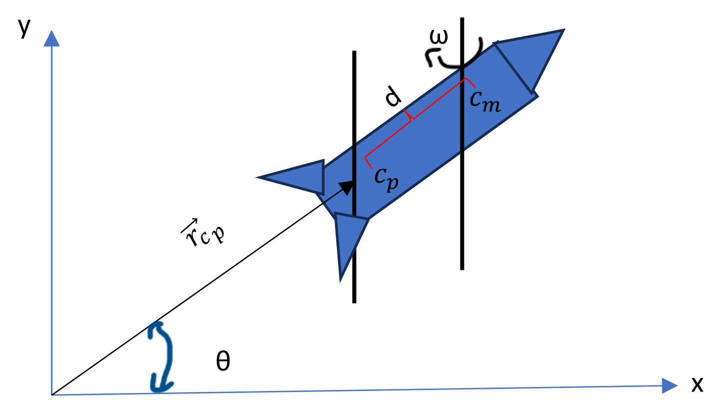

Figure1.Free body diagram for the model

-We will compress the rocket in a point like particle with all of its shape properties embedded in a point c_p, called pressure center where all dynamics will be applied. To take into account the shape and the rotational motion, described the constant angular frequency ω around the center of mass, caused by the shape we will use the parallel axis theorem to express the moment of inertia along the pressure center with respect to the center of mass c_m, and the separation distance d between those points.

-The last trajectory approximation will be implemented by using polar coordinates where, $\overrightarrow{r_{c_p}}$ represents the position vector from origin to the pressure center and θ being the polar angle.

Equations of motion:

$$
\begin{cases}
m(t) = m_r(t) + m_f(t) \rightarrow \begin{cases} m_r(t) \rightarrow mass\ of\ the\ dry\ rocket (2\ modules) \\ m_f(t) \rightarrow fuel\ mass \end{cases} \\
D = \frac{1}{2}\rho_h C_d A v^2(t) \rightarrow \begin{cases} \rho_h \rightarrow altitude\ dependent\ air\ density \\ C_d \rightarrow drag\ coefficent \\ A \rightarrow cross\ sectional\ area\ of\ the\ rocket \\ v(t) \rightarrow rocket\ velocity \end{cases} \rightarrow drag\ force \\
Tr = \dot{m_f}(t)\sqrt{\frac{2(p_c - p_e)}{\rho}} + (p_c - p_a)A_e \rightarrow \begin{cases} p_c \rightarrow chamber\ pressure\ of\ the\ fuel \\ p_e \rightarrow exausht\ pressure\ of\ the\ fuel \\ \rho \rightarrow fuel\ density \\ p_a \rightarrow ambient\ pressure \\ A_e \rightarrow exit\ area\ of\ the\ nozzle \end{cases} \rightarrow thrust\ force \\
\dot{m_f}(t) = \rho\frac{\pi r^4}{8\mu L}(p_c - p_e) + f_{non-linear\ effects}\big(m_f(t), v(t)\big) \rightarrow \begin{cases} r \rightarrow effective\ radius\ of\ the\ flow\ chanel \\ L \rightarrow nozzle\ throat \\ \mu \rightarrow dynamic\ viscosity\ of\ the\ fuel \end{cases} \\
R = \frac{\rho v(t) L}{\mu} \rightarrow Reynolds\ number < 2300 \rightarrow Laminar\ flow\ condition
\end{cases}
$$

Key parameters

-Utilizing the Euler-Lagrange formalism with nonconservative forces, thrust and drag, we obtain the following set of differential equations which will describe the dynamics of the system:

$$
\begin{cases}
\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = \frac{\partial Tr}{\partial \dot{q}} - \frac{\partial D}{\partial \dot{q}}, q \in \left\{r_{c_p}, \theta\right\} \rightarrow Euler - Lagrange\ equations \\
L = \frac{m(t)}{2}\left(\dot{r_{c_p}}^2 + r_{c_p}^2\dot{\theta}^2\right) + \frac{\omega^2}{2}\left(I_{cm}(r_{c_p}) + m(t)d(r_{c_p})^2\right) - mgr_{c_p}\sin(\theta) \rightarrow Lagrangian \\
\begin{cases}
\frac{d}{dt}\left(m(t)\dot{r_{c_p}}\right) - m(t)r_{c_p}\dot{\theta}^2 + mg\sin(\theta) + \frac{\omega^2}{2}\left(\frac{\partial I_{cm}(r_{c_p})}{\partial r_{c_p}} + 2m(t)d(r_{c_p})\frac{\partial d(r_{c_p})}{\partial r_{c_p}}\right) = \frac{\partial f_{non-linear\ effects}}{\partial \dot{r_{c_p}}} - \rho_h C_d \dot{r_{c_p}} \rightarrow radial\ equation \\
\frac{d}{dt}\left(m(t)r_{c_p}\dot{\theta}\right) + mgr_{c_p}\cos(\theta) = \frac{\partial f_{non-linear\ effects}}{\partial \dot{\theta}} - \rho_h C_d r_{c_p}^2\dot{\theta} \rightarrow angular\ equation
\end{cases}
\end{cases}
$$

Future aspects and perspectives:

-For a defined form of $f_{non-linear\ effects}$ we will be able to solve the differential equations, either analytically or numerically, and be able to obtain m(t) and v(t) for our system. The main goal is to ensure that m(t) and v(t) are parametrized in such a way that the model allows a realistic estimation of the necessary parameters which will guide the rocket to h = 9000 m.

-Further adjustments with the experiments and simulations are in progress in hope of a good symbiosis between theory and experiment.